Very important set of MCQ, including repeating exam points and values. Will also serve as a practice set(Answers at the end)—----------by HMHDental

BDS Professional Examination: Embryology (Weeks 1–8)

1. A 23-year-old female presents with acute pelvic pain. Which of the following is the most frequent site for an ectopic implantation?

a. Internal os

b. Rectouterine pouch

c. Ampulla of the uterine tube

d. Isthmus of the uterine tube

e. Ovary

2. Human Chorionic Gonadotropin (hCG) produced by the syncytiotrophoblast can first be detected in maternal blood on:

a. Day 2

b. Day 6

c. Day 8

d. Day 12

e. Day 14

3. In clinical practice, a positive pregnancy test using maternal urine is typically achievable by:

a. Day 2

b. Day 6

c. Day 10

d. Day 14

e. Day 21

4. The erosion of maternal sinusoids and the subsequent secretion of hCG into the maternal circulation is the function of the:

a. Epiblast

b. Hypoblast

c. Cytotrophoblast

d. Embryoblast

e. Syncytiotrophoblast

5. During the second week of development, the inner cell mass (embryoblast) differentiates into the:

a. Ectoderm and Endoderm

b. Epiblast and Hypoblast

c. Syncytiotrophoblast and Cytotrophoblast

d. Somatopleure and Splanchnopleure

e. Trophoblast and Blastocoel

6. Implantation in the lower uterine segment, potentially obstructing the internal os, is clinically defined as:

a. Ectopic pregnancy

b. Abdominal pregnancy

c. Lithopedion

d. Placenta previa

e. Hydatidiform mole

7. The critical process of gastrulation in the third week is initiated by the formation of the:

a. Neural tube

b. Notochord

c. Prechordal plate

d. Primitive streak

e. Prochordal plate

8. A neonate presents with a sacrococcygeal tumor containing derivatives of all three germ layers. This pathology arises from remnants of the:

a. Neural crest

b. Prochordal plate

c. Notochord

d. Primitive streak

e. Oropharyngeal membrane

9. Statistically, the most common site for an ectopic pregnancy is the:

a. Ovary

b. Ampulla

c. Cervix

d. Abdominal cavity

e. Broad ligament

10. The closure of the cranial (anterior) neuropore occurs on approximately:

a. Day 22

b. Day 24

c. Day 25

d. Day 28

e. Day 30

11. A failure of the neural folds to fuse in the cranial region results in:

a. Anencephaly

b. Spina bifida

c. Microcephaly

d. Encephalocele

e. Hydrocephalus

12. The incidence of Neural Tube Defects (NTDs) is significantly reduced by the periconceptional administration of:

a. Retinoic acid

b. Ascorbic acid

c. Cholecalciferol

d. Folic acid

e. Ferrous sulfate

13. Odontoblasts, which form the dentin of teeth, are derived from which embryonic cell population?

a. Surface ectoderm

b. Neural crest cells

c. Paraxial mesoderm

d. Lateral plate mesoderm

e. Endoderm

14. Underdevelopment of the zygomatic bones and mandibular hypoplasia (Treacher Collins Syndrome) is associated with defects in:

a. Neural tube closure

b. Neural crest cell migration

c. Splanchnic mesoderm

d. Endodermal pouching

e. Ectodermal placodes

15. The ampulla of the uterine tube is most significant as the site for:

a. Implantation

b. Fertilization

c. Cleavage

d. Morula formation

e. Capacitation

16. An infant is born with abdominal viscera herniating through an enlarged umbilical ring, covered by a sac of amnion. This is:

a. Gastroschisis

b. Ectopia cordis

c. Omphalocele

d. Diaphragmatic hernia

e. Bladder exstrophy

17. Failure of the lateral body folds to fuse in the thoracic region leads to the heart protruding outside the chest, known as:

a. Gastroschisis

b. Omphalocele

c. Anencephaly

d. Ectopia cordis

e. Bifid ribs

18. A ventral body wall defect characterized by the protrusion of bowel loops directly into the amniotic fluid (not covered by a sac) is:

a. Umbilical hernia

b. Omphalocele

c. Gastroschisis

d. Hirschsprung disease

e. Cloacal exstrophy

19. The embryonic period, where the embryo is most susceptible to gross structural malformations from teratogens, occurs during:

a. Weeks 1–2

b. Weeks 2–3

c. Weeks 3–8

d. Weeks 9–12

e. Weeks 13–38

20. A child born with limb reduction defects such as Phocomelia was likely exposed in utero to:

a. Ethanol

b. Nicotine

c. Warfarin

d. Thalidomide

e. Tetracycline

21. Neurulation is the process by which the neural plate forms the neural tube, a process induced by the:

a. Primitive streak

b. Prochordal plate

c. Notochord

d. Neural crest

e. Paraxial mesoderm

22. The cavity that develops within the epiblast and eventually surrounds the entire embryo is the:

a. Yolk sac

b. Blastocyst cavity

c. Extraembryonic coelom

d. Chorionic cavity

e. Amniotic cavity

23. Congenital megacolon (Hirschsprung disease) results from a failure of migration of:

a. Paraxial mesoderm

b. Neural crest cells

c. Endoderm

d. Surface ectoderm

e. Sclerotome

24. The primary and secondary umbilical vesicles (yolk sacs) are lined by:

a. Ectoderm

b. Endoderm

c. Mesoderm

d. Trophoblast

e. Epiblast

25. Exposure to teratogenic agents during the 3rd to 8th weeks of gestation primarily results in:

a. Minor functional disturbances

b. Major structural anomalies

c. Physiological delays

d. No observable effect

e. Immediate fetal death

26. The structure that attaches the embryo to the trophoblastic wall and later contributes to the umbilical cord is the:

a. Connecting stalk

b. Placenta

c. Amnion

d. Allantois

e. Yolk sac

27. By the end of the 5th week of development, the total number of somite pairs is approximately:

a. 20–22

b. 30–32

c. 42–44

d. 10–12

e. 50–52

28. The primary organizer of the cranial region of the embryo is the:

a. Primitive node

b. Prechordal plate

c. Cloacal membrane

d. Oropharyngeal membrane

e. Notochord

29. A trophoblastic tumor, such as a hydatidiform mole, is clinically characterized by excessively high levels of:

a. Alpha-fetoprotein

b. Estradiol

c. Progesterone

d. hCG

e. Luteinizing hormone

30. The chorionic cavity is the clinical term for the:

a. Amniotic cavity

b. Primary yolk sac

c. Blastocyst cavity

d. Extraembryonic coelom

e. Intraembryonic coelom

31. In the 3rd week, the first sign of the beginning of gastrulation is the appearance of the:

a. Neural groove

b. Notochord

c. Primitive streak

d. Somites

e. Heart bulge

32. The integumentary system (epidermis) and the nervous system are derivatives of the:

a. Endoderm

b. Mesoderm

c. Ectoderm

d. Neural crest

e. Trophoblast

33. The epithelial lining of the gastrointestinal and respiratory systems is derived from:

a. Ectoderm

b. Mesoderm

c. Endoderm

d. Epiblast

e. Hypoblast

34. Rhythmic contractions of the primitive heart tube (heart beat) begin on:

a. Day 18

b. Day 20

c. Day 22

d. Day 25

e. Day 28

35. The formation of blood and blood vessels (vasculogenesis) begins in the extraembryonic mesoderm during:

a. Week 1

b. Week 2

c. Week 3

d. Week 4

e. Week 5

36. Communication between the primitive oral cavity (stomodeum) and the pharynx is established by the rupture of the:

a. Cloacal membrane

b. Oropharyngeal membrane

c. Pleuropericardial membrane

d. Diaphragm

e. Primary palate

37. A patient with Spina Bifida Occulta typically presents with which clinical sign?

a. Large meningeal sac

b. Exposed spinal cord

c. Tuft of hair in the lumbar region

d. Paralysis of lower limbs

e. Hydrocephalus

38. The sequence of somite formation in the embryo follows a pattern that is:

a. Caudocranial

b. Craniocaudal

c. Mediolateral

d. Lateromedial

e. Random

39. Skeletal muscle and the vertebrae are derived from the:

a. Intermediate mesoderm

b. Lateral plate mesoderm

c. Paraxial mesoderm

d. Endoderm

e. Ectoderm

40. The transition from the embryonic period to the fetal period occurs at the end of:

a. Week 4

b. Week 6

c. Week 7

d. Week 8

e. Week 12

41. The development of the forebrain is induced by the underlying:

a. Notochord

b. Prechordal plate

c. Primitive streak

d. Heart tube

e. Cloacal membrane

42. The second week of development is frequently referred to as the "Week of Twos" because of the formation of the:

a. Trilaminar disc

b. Neural tube

c. Bilaminar disc

d. Somites

e. Body folds

43. Which of the following is the source of the extraembryonic mesoderm?

a. Epiblast

b. Hypoblast

c. Cytotrophoblast

d. Yolk sac endoderm

e. Syncytiotrophoblast

44. The pharyngeal (branchial) arches, essential for head and neck development, begin to appear during:

a. Week 2

b. Week 3

c. Week 4

d. Week 6

e. Week 8

45. After serving its role in gastrulation, the primitive streak normally regresses and disappears by the end of:

a. Week 2

b. Week 3

c. Week 4

d. Week 5

e. Week 8

46. Which of the following is a derivative of the neural tube rather than neural crest cells?

a. Melanocytes

b. Motor neurons of the spinal cord

c. Odontoblasts

d. Adrenal medulla

e. Schwann cells

47. The differentiation of the trophoblast into two layers and the embryoblast into two layers is a hallmark of:

a. Week 2

b. Week 3

c. Week 4

d. Week 5

e. Week 6

48. Hemorrhage from an ectopic pregnancy is most life-threatening when implantation occurs in the:

a. Ampulla

b. Isthmus

c. Intramural part

d. Ovary

e. Abdominal cavity

49. The amniotic cavity begins its development as a small space within the:

a. Epiblast

b. Hypoblast

c. Cytotrophoblast

d. Syncytiotrophoblast

e. Extraembryonic mesoderm

50. The secondary umbilical vesicle (yolk sac) is smaller than the primary yolk sac and is formed within the:

a. Epiblast

b. Hypoblast

c. Exocoelomic cavity

d. Amniotic cavity

e. Primitive streak

51. The formation of the neural plate is induced by the:

a. Primitive streak

b. Notochord

c. Neural crest

d. Somites

e. Allantois

52. The neuroectoderm is the embryonic precursor to the:

a. Surface epidermis

b. Central nervous system

c. Dermis of the skin

d. Smooth muscle

e. Liver parenchyma

53. Upper limb buds become recognizable on the ventrolateral body wall during:

a. Week 2

b. Week 3

c. Week 4

d. Week 6

e. Week 8

54. Folding of the embryo, transforming the flat disc into a 3D internal structure, is caused by:

a. Gastrulation

b. Neurulation

c. Rapid growth of the embryonic disc

d. Regression of the yolk sac

e. Formation of the placenta

55. At the 8th week of development, the head constitutes approximately what portion of the Crown-Rump Length?

a. One-fourth

b. One-third

c. One-half

d. One-fifth

e. One-tenth

56. Anencephaly is a lethal condition caused by the failure of closure of the:

a. Caudal neuropore

b. Cranial neuropore

c. Oropharyngeal membrane

d. Pleuroperitoneal folds

e. Primitive streak

57. Before the establishment of the uteroplacental circulation, the embryo is nourished by:

a. Maternal blood

b. Yolk proteins

c. Uterine gland secretions

d. Amniotic fluid

e. Phagocytosis

58. In a normal pregnancy, the site where the sperm and oocyte meet is the:

a. Uterine cavity

b. Ampulla of the uterine tube

c. Isthmus of the uterine tube

d. Cervical canal

e. Infundibulum

59. The morula stage, consisting of approximately 12 to 16 blastomeres, is reached at:

a. 12 hours post-fertilization

b. 24 hours post-fertilization

c. 48 hours post-fertilization

d. 3 days post-fertilization

e. 6 days post-fertilization

60. For implantation to occur, the blastocyst must first shed its:

a. Corona radiata

b. Plasma membrane

c. Zona pellucida

d. Hypoblast

e. Syncytiotrophoblast

61. Attachment of the blastocyst to the endometrial epithelium (implantation) typically begins on:

a. Day 2

b. Day 4

c. Day 6

d. Day 10

e. Day 14

62. The epithelial lining of the bladder and urethra is a derivative of the:

a. Ectoderm

b. Mesoderm

c. Endoderm

d. Neural crest

e. Epiblast

63. The conversion of endometrial cells into secretory, glycogen-rich cells at the site of implantation is the:

a. Acrosome reaction

b. Decidual reaction

c. Cortical reaction

d. Zona reaction

e. Capacitation

64. A primary chorionic villus is structurally defined as having:

a. Syncytiotrophoblast only

b. A cytotrophoblastic core covered by syncytiotrophoblast

c. A mesodermal core

d. Fetal capillaries

e. Maternal blood vessels

65. A chorionic villus that has developed a core of extraembryonic mesoderm is a:

a. Primary villus

b. Secondary villus

c. Tertiary villus

d. Stem villus

e. Anchoring villus

66. The presence of fetal capillaries within the mesodermal core of a chorionic villus defines it as a:

a. Primary villus

b. Secondary villus

c. Tertiary villus

d. Decidual villus

e. Anchoring villus

67. In cases of abdominal pregnancy, the most frequent site of implantation is the:

a. Liver

b. Greater omentum

c. Rectouterine pouch

d. Spleen

e. Mesentery

68. The appearance of lacunae in the syncytiotrophoblast, which later fill with maternal blood, occurs during:

a. Week 1

b. Week 2

c. Week 3

d. Week 4

e. Week 5

69. The oropharyngeal membrane is a bilaminar structure composed of:

a. Ectoderm and Mesoderm

b. Endoderm and Mesoderm

c. Ectoderm and Endoderm

d. Ectoderm only

e. Endoderm only

70. The somite is divided into three components; the one responsible for the formation of the vertebral column is the:

a. Sclerotome

b. Myotome

c. Dermatome

d. Epimere

e. Hypomere

71. The first organ system to reach a functional state in the developing embryo is the:

a. Central nervous system

b. Respiratory system

c. Cardiovascular system

d. Urinary system

e. Gastrointestinal system

72. The epiblast is the source of which of the following germ layers?

a. Ectoderm

b. Mesoderm

c. Endoderm

d. All of the above

e. None of the above

73. Sirenomelia, a defect involving the fusion of the lower limbs, is also known as:

a. Mermaid syndrome

b. Lobster claw deformity

c. Phocomelia

d. Amelia

e. Polydactyly

74. A severe neural tube defect characterized by an open brain and spinal cord is:

a. Craniorachischisis

b. Anencephaly

c. Spina bifida cystica

d. Encephalocele

e. Microcephaly

75. The "Embryonic Period" (organogenesis) spans from:

a. Weeks 1 to 2

b. Weeks 3 to 8

c. Weeks 9 to 12

d. Weeks 13 to 20

e. Weeks 21 to 38

76. The onset of the heartbeat in the embryo occurs approximately on:

a. Day 18

b. Day 21

c. Day 24

d. Day 28

e. Day 32

77. The ventral displacement of the connecting stalk occurs as a result of:

a. Cephalic folding

b. Tail folding

c. Lateral folding

d. Gastrulation

e. Neurulation

78. The first morphological evidence of the nervous system is the neural plate, appearing on:

a. Day 15

b. Day 18

c. Day 21

d. Day 25

e. Day 28

79. During the 4th and 5th weeks, the age of the embryo is most accurately determined by counting the:

a. Villi

b. Somites

c. Branchial arches

d. Limb buds

e. Neural folds

80. Cleft lip results from a failure of fusion between the:

a. Maxillary and medial nasal processes

b. Maxillary and lateral nasal processes

c. Medial and lateral nasal processes

d. Mandibular processes

e. Frontonasal and maxillary processes

Answer Key

Q 1-8	Q 9-16	Q 17-24	Q 25-32	Q 33-40	Q 41-48	Q 49-56	Q 57-64	Q 65-72	Q 73-80
c	b	d	b	c	b	a	c	b	a
c	c	c	a	c	c	b	b	c	a
b	a	c	c	c	d	c	d	c	a
e	d	e	b	b	c	c	c	a	a

d	**b**	**d**	**a**	**b**	**e**	**a**	**c**	**d**	**b**
e	**a**	**c**	**b**	**b**	**d**	**d**	**c**	**e**	**b**
a	**b**	**e**	**a**	**b**	**a**	**a**	**b**	**a**	**d**
d	**c**	**d**	**c**	**d**	**e**	**e**	**b**	**e**	**a**